

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2020

Subject: 353152 Electronic Circuits I

Section: 5-6

Date: 3 September 2020

Time: 13.00-15.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 2 sections. There will be 13 pages (including this page) and they are worth 89 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Part 1 (64 points) Multiple choices

Instruction: Mark the correct answer for the following questions in the provided answer sheet.

1. Conductors are atoms that contain _____ valence electrons.
 (a) 2 (b) 4 (c) 6 (d) 8

2. Which type of material has narrowest energy gap between conduction band and valence band?
 (a) Insulator (b) Semiconductor
 (c) Conductor (d) Resistor

3. Atoms of semiconductors contain _____ valence electrons.
 (a) 2 (b) 4 (c) 6 (d) 8

4. What is the voltage drop over Germanium diode when it is in forward bias mode?
 (a) 0.2 V (b) 0.3 V (c) 0.6 V (d) 0.7 V

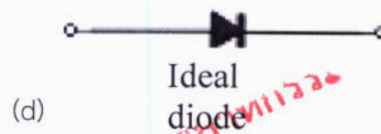
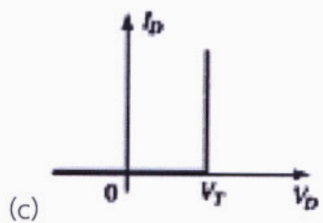
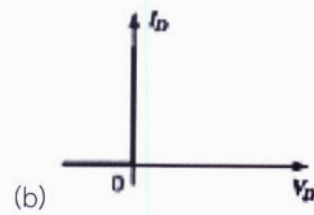
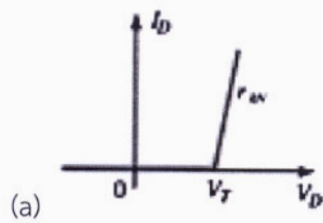
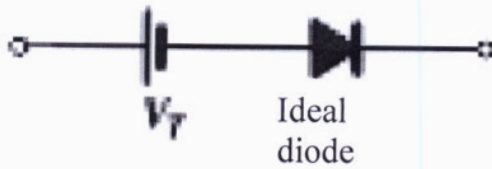
5. Which one of the following is true?
 (a) Diode is open circuit when operating in reverse bias mode.
 (b) Forward bias means the voltage on p junction is lower than n junction.
 (c) For a practical diode, there is no voltage drop for forward bias mode.
 (d) The voltage drops of a Silicon diode are at 0.3 V when they are operating in forward bias mode.

6. Which circuit provides output as $V_{DC} = 0.636(V_m - V_k)$ for practical diode?
 (a) Half Wave Rectifier (b) Bridge Rectifier
 (c) Positive Half-Wave Rectifier (d) Center-Tapped Transformer Rectifier

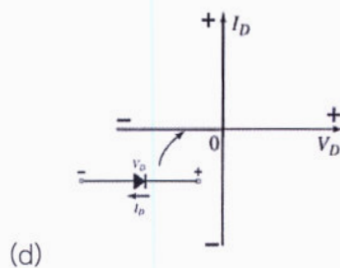
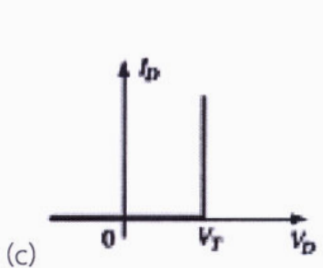
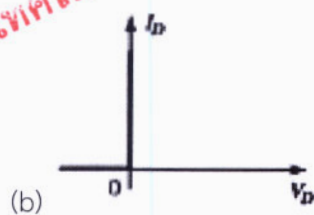
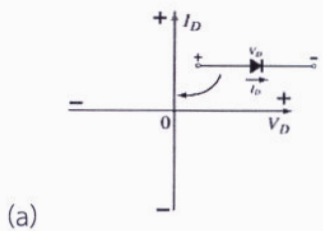
7. Which one of the following is NOT true?
 (a) For practical diode, the bridge-rectifier has lower voltage drop at diode compare with center-tapped transformer rectifier.
 (b) The cost of bridge-rectifier is higher than the cost of half-wave rectifier.
 (c) Half-wave rectifier provides higher dc voltage output than full-wave rectifier.

- (d) Bridge-rectifier contains more number of diodes over center-tapped transformer rectifier.
8. Which of the following transistor operating mode represents the open circuit?
(a) Cut-off (b) Saturation (c) Active (d) None
9. Which of the following is true for common base configuration?
(a) BE junction is reverse biased when transistor is in cutoff mode.
(b) CE junction is forward biased when transistor is in saturation mode.
(c) CB Junction is reverse biased when transistor is in saturation mode.
(d) CB Junction is forward biased when transistor is in cutoff mode.
10. Which type of diode has operated reverse voltage between 1.8 to 200 V?
(a) Silicon Diode (b) Germanium Diode
(c) Zener Diode (d) Light Emitting Diode
11. When the transistor is operating in saturation mode, what is the V_{CE} approximately be?
(a) 0 V (b) V_{CC} (c) 0.7 V (d) ∞
12. The intersection of the load line with the characteristic curve determines the _____ of the system.
(a) quiescent point (b) load-line analysis
(c) characteristic curve (d) forward bias
13. When the transistor is operating in cutoff mode, what is the I_C approximately?
(a) 0 A (b) I_B (c) I_{CQ} (d) ∞
14. How much voltage drop over silicon diode when it is in forward bias mode?
(a) 0.2 V (b) 0.3 V (c) 0.6 V (d) 0.7 V

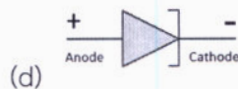
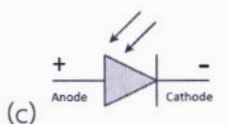
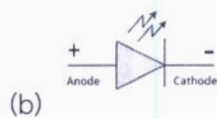
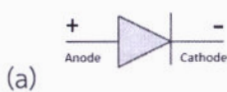
15. Which one of the answers is corresponded to following figure?



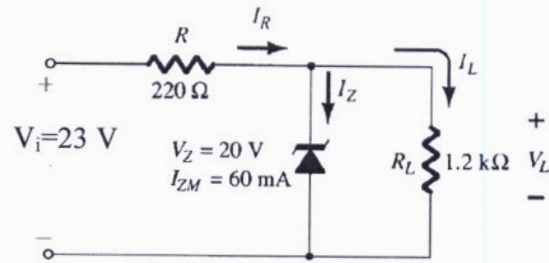
16. The ideal diode acts as an open switch. Which one is the diode characteristic?



17. Which one of the answers is the symbol of light emitting diode?

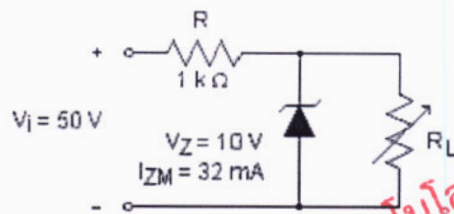


18. Determine V_L .



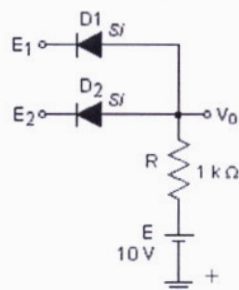
- (a) 23 V (b) 0 V (c) 3.5 V (d) 20 V

19. With this zener diode in its "on state," what is the level of I_Z (current of zener diode) for the maximum load resistance?



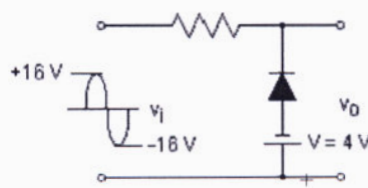
- (a) 0 mA (b) $I_Z = 8\text{mA}$ (c) $I_Z = 40\text{mA}$ (d) $I_Z = 32\text{mA}$

20. Determine the current through each diode if $E_1 = E_2 = 0\text{ V}$.



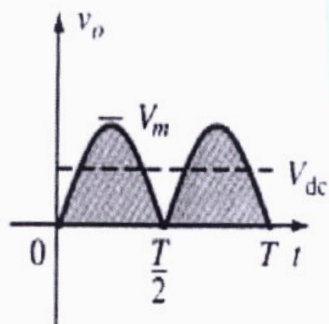
- (a) 4.65 mA (b) 9.3 mA (c) 18.68 mA (d) 0.7 mA

21. Determine the peak for both half cycles of the output waveform. Assume that the diode is ideal.



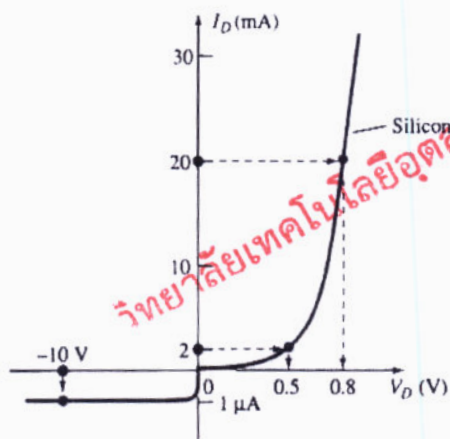
- (a) 16 V, -4 V (b) 16 V, 4 V (c) -16 V, 4 V (d) -16 V, -4 V

22. Which of the following is the average output value of full-wave rectifier?



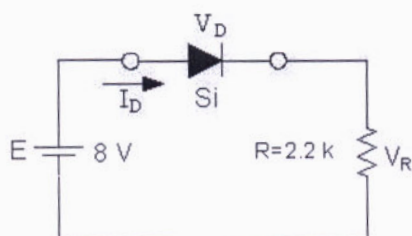
- (a) $0.318V_m$ (b) $0.5V_m$ (c) $0.636V_m$ (d) $1.414V_m$

23. From figure below, what is the DC resistance at 20 mA?



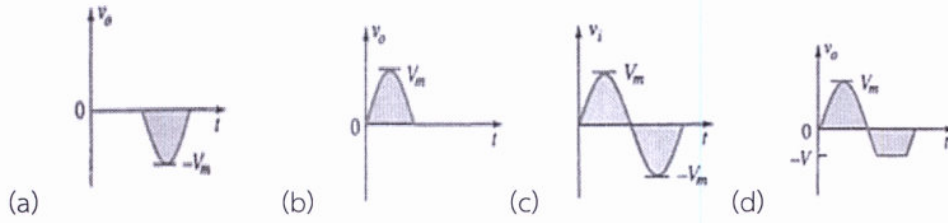
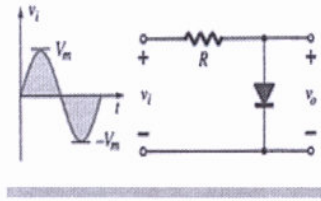
- (a) $4 \text{ m}\Omega$ (b) 4Ω (c) 40Ω (d) $40 \text{ k}\Omega$

24. Find V_d , and V_R

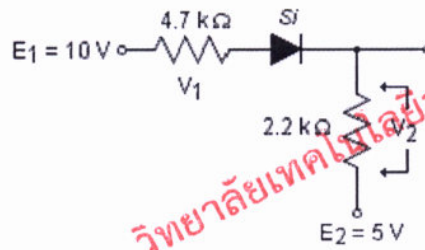


- (a) 0.7V and 7.3V (b) 0.3V and 7.7V (c) 0.7V and 0.3V (d) 0.3 V and 8V

25. What is the output voltage of the ideal diode circuit as shown in figure below?

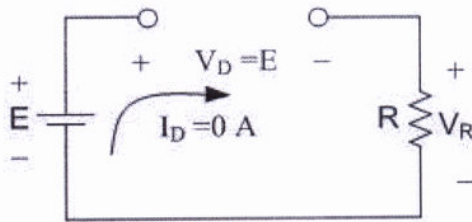


26. Determine V_2 .



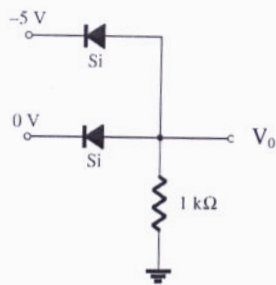
- (a) 3.201 V (b) 0 V (c) 4.3 V (d) 1.371 V

27. Find V_R where the given $E=20V$.



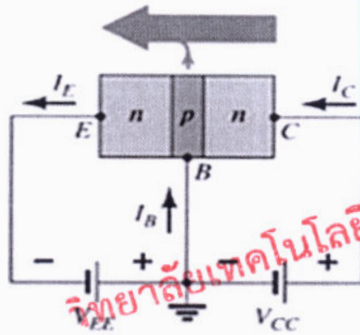
- (a) 0.7V
(b) 0.3V
(c) 0 V
(d) 20V

28. Determine V_0 from ideal diode circuit below.



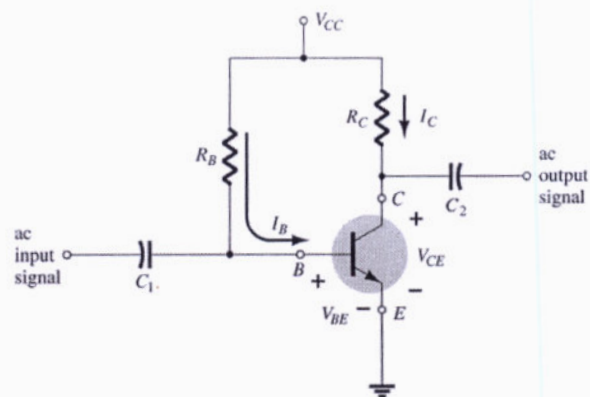
- (a) -5 V (b) -4.3 V (c) 0.7 V (d) 1.4 V

29. What is a transistor configuration of the circuit in figure below?



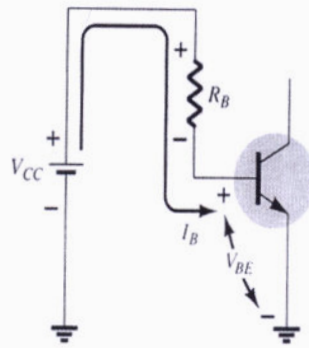
- (a) Common Emitter (b) Common Collector
(c) Common Base (d) Common Emitter feedback

30. What is a bias type of the circuit in figure below?



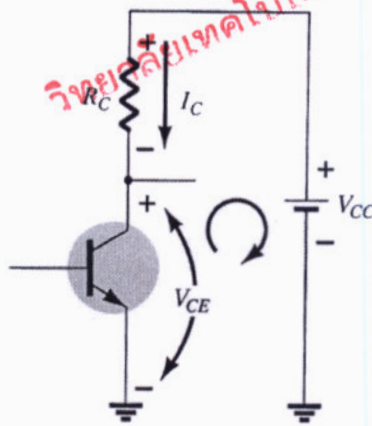
- (a) Fixed-bias circuit (b) Emitter-stabilized bias circuit
(c) Voltage divider bias circuit (d) DC bias with voltage feedback

31. From Base-Emitter loop in picture below, what is the Kirchhoff's voltage equation for the loop?



- (a) $V_{CC} - I_B R_B - V_{BE} = 0$ (b) $V_{CC} - I_B R_B + V_{BE} = 0$
 (c) $V_{CC} + I_B R_B - V_{BE} = 0$ (d) $V_{CC} + I_B R_B + V_{BE} = 10$

32. From Collector-Emitter loop in picture below, what is the Kirchhoff's voltage equation for the loop?

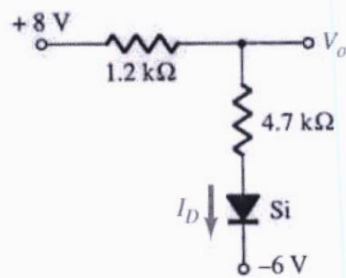


- (a) $V_{CE} + I_C R_C + V_{CC} = 0$ (b) $V_{CE} - I_C R_C + V_{CC} = 0$
 (c) $V_{CC} - I_B R_B - V_{BE} = 0$ (d) $V_{CE} + I_C R_C - V_{CC} = 0$

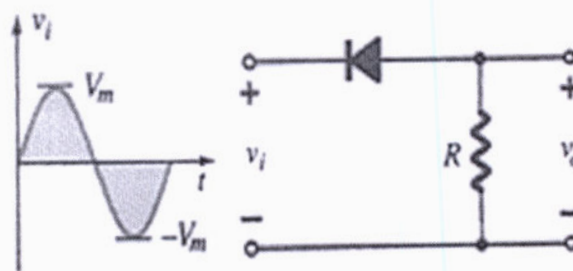
Part 2 (25 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

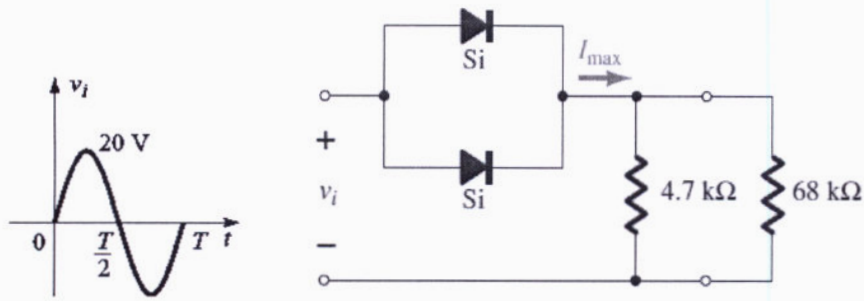
1. Determine I_D and V_o from figure below



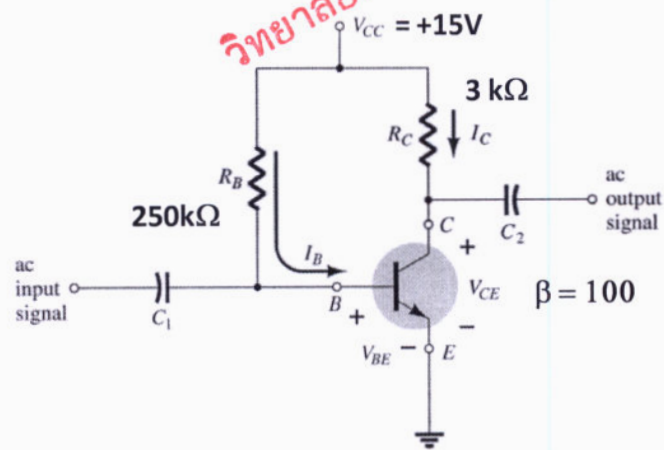
2. Sketch the output waveform of the network below. Given $V_m=5V$. Assume that the diode is ideal.



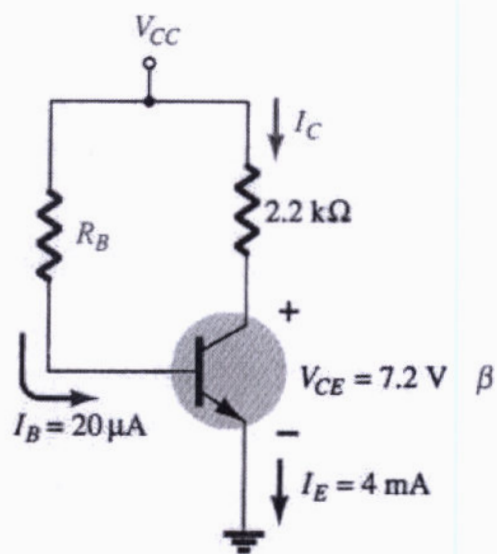
3. From the circuit, sketch the output and determine dc output voltage.



4. Determine I_B and I_C of the circuit below.



5. Determine I_C , β , R_B and V_{CC} of the circuit below.



Answer Sheet

Name: _____ ID: _____

Subject: 353152 Electronic Circuits I

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